LAB 10: Ordinary Differential Equation

Objective: To numerically obtain the solution of generic governing ODE of any physical system

Theory: equations, which are composed of an unknown function and its derivatives, are called differential equations. When the function involves one independent variable, the equation is called an *ordinary differential equation* (or ODE). Consider an ordinary differential equations of the form,

$$\frac{dy}{dt} = f(t, y)$$

Euler's Method –

New value = old value + slope × step size

in mathematical terms,

$$y_{i+1} = y_i + \phi h$$

where the slope ϕ is called an increment function. According to this equation, the slope estimate of ϕ is used to extrapolate from an old value yi to a new value yi+1 over a distance h. This formula can be applied step by step to trace out the trajectory of the solution into the future.

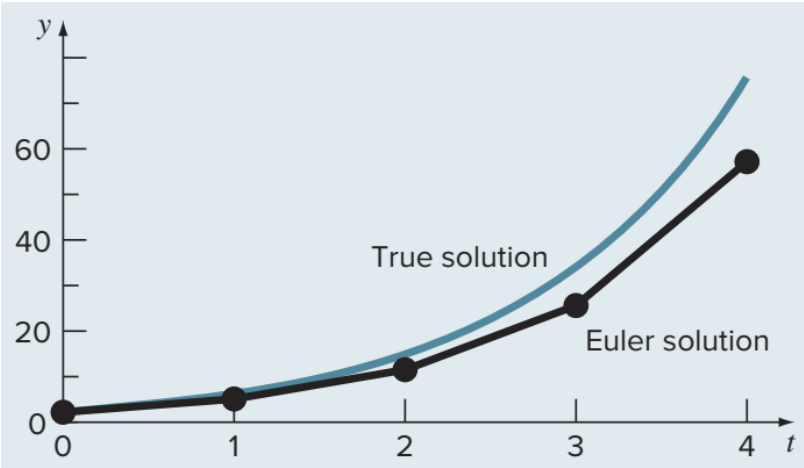
The first derivative provides a direct estimate of the slope at t_i

$\phi = f(t_i, y_i)$
where $f(t_i, y_i)$ is the differential equation evaluated at t_i and y_i.
$y_{i+1} = y_i + f(t_i, y_i)h$
This formula is referred to as *Euler's method* (or the Euler-Cauchy or point-slope method). A new value of y is predicted using the slope (equal to the first derivative at the original value of t) to extrapolate linearly over the step size h.

Problem Statement. Use Euler's method to integrate $y' = 4e^{0.8t} - 0.5y$ from $t = 0$ to 4 with a step size of 1. The initial condition at $t = 0$ is $y = 2$. Note that the exact solution can be determined analytically as

$$y = \frac{4}{1.3}(e^{0.8t} - e^{-0.5t}) + 2e^{-0.5t}$$

t	y_{true}	y_{Euler}	$\lvert\varepsilon_t\rvert$ (%)
0	2.00000	2.00000	
1	6.19463	5.00000	19.28
2	14.84392	11.40216	23.19
3	33.67717	25.51321	24.24
4	75.33896	56.84931	24.54



Assignment Problem –

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Read about Modified Euler method and Attempt a numerical based on it.

One numerical as suggested by Faculty.

Sample numerical.

Solve the following initial value problem over the interval from $t = 0$ to where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dt} = yt^2 - 1.1y$$

(a) Analytically.
(b) Using Euler's method with $h = 0.5$ and 0.25.

Modified euler method:

→ The regular method uses the slope only a the beginning of the interval. Modified euler improves accuracy by averaging the slope at the start and end of each step.

formula:

$$y_{i+1} = y_i + f(t_i, y_i) \cdot h \quad \rightarrow \text{Step-1 Predictor}$$

$$y_{i+1} = y_i + \frac{f(t_i, y_i) + f(t_{i+1}, y_{i+1})}{2} \cdot h$$

Working:

- Predicts: takes rough step using current slope
- Evaluate: find slope at the predicted point.
- Correct: Average both slope and take a better step.

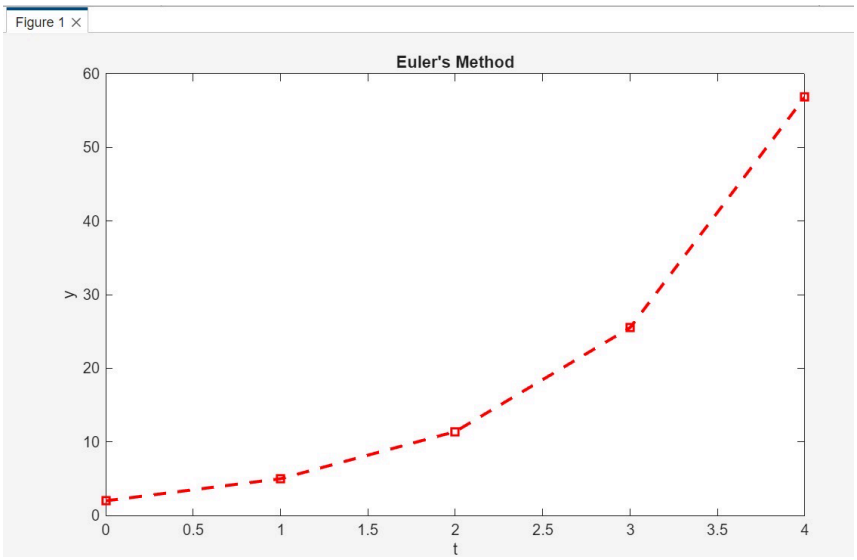
Solving via simplified euler method:

Code:

exp10cel.m

:\Users\setud\OneDrive\Documents\MATLAB\exp10cel.m

```
% LAB 10: Euler's Method for ODE
% y' = 4e^(0.8t) - 0.5y, y(0) = 2, h = 1, t = 0 to 4

h = 1;
t = 0;
y = 2;

fprintf(' t \t y_euler\n');

fprintf(' %d \t %.5f\n', t, y);

t_vals = t;
y_vals = y;

for t = 0:h:3
    y = y + (4*exp(0.8*t) - 0.5*y) * h;
    t_vals = [t_vals, t+h];
    y_vals = [y_vals, y];
    fprintf(' %d \t %.5f\n', t+h, y);
end

plot(t_vals, y_vals, 'r--s', 'LineWidth', 2);
xlabel('t');
ylabel('y');
title("Euler's Method");
```

Result:

```
>> exp10cel
 t        y_euler
 0        2.00000
 1        5.00000
 2        11.40216
 3        25.51321
 4        56.84931
```